Building a Multi-Style "Barbell" Factor Strategy in China's A-Share Market

Today’s Agenda

90 minutes | Introductory Session

Part A

Overview of Quantitative Investment

30 min

Part B

Project Brief & Technical Framework

40 min

Part C

Team Formation & Next Steps

20 min

A

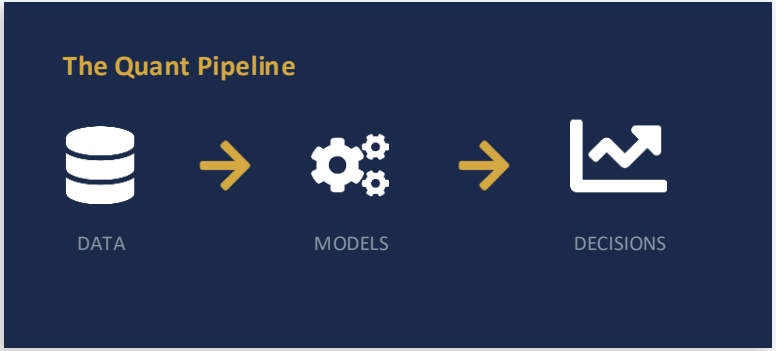
Overview of Quantitative Investment

What it is, why it matters, and how it works in China’s A-share market

What Is Quantitative Investing?

Systematic, rules-based investment using data and models to make decisions

Instead of relying on subjective judgment, quantitative investing uses mathematical models, statistical analysis, and computational techniques to identify opportunities and manage portfolios.



Discretionary / Fundamental

- Analyst intuition & judgment
- Deep-dive on few companies
- Subjective, hard to replicate

Quantitative / Systematic

- Data-driven models & rules
- Broad coverage (1000s of stocks)
- Repeatable, scalable, testable

Quant Strategy Landscape: Key Dimensions

5 Dimensions of Quant Strategies

Asset Universe: Equities, bonds, futures (commodity, index), FX, options, other

Signal Sources: Fundamental (value, quality, growth) or technical (price, volume)

Holding Period: Intraday; short (days); medium (1–3M); long (4–6M+)

Position Style: Long-only vs. market-neutral / hedged

Strategy Type: Stock selection, sector rotation, macro, timing, stat arb, trend-following

Where This Project Focuses

1. **Asset Universe**
   A-share equities (CSI All-Share)
2. **Signal Sources**
   Financial statement data + market data (price, volume)
3. **Holding Period**
   Medium-term: monthly rebalance (1–3 month horizon)
4. **Strategy Types**
   Smart Beta

Factors & Investment Logic: An Analogy

Apples = Stocks | Their attributes = Factors

"All models are approximations of reality."

No.	Variety	Wt.(kg)	Origin	Color	Sweet ness	Price(¥)
1	Fuji	0.5	Xinjiang	Red	10	20
2	Granny Smith	0.4	Shandong	Green	5	15
3	Fuji	0.3	Shaanxi	Red-Yellow	8	12
4	Golden Delicious	0.3	Liaoning	Gold	6	13
5	Red Delicious	0.4	Aus.	Red	4	18
6	Red Delicious	0.2	N.Am.	Yellow	4	16
...						

** Illustrative only, not real market data*

Evolution of Factor Investing

From single-factor CAPM to multi-factor AI-driven systems: 60 years of pricing evolution

1960s
CAPM
Sharpe, Lintner: Market beta (β) as sole risk factor

1976
APT
Ross: Multiple macro factors drive returns

1981–92
Anomalies
Banz (Size), Basu (P/E), early Value evidence

1993
FF 3-Factor
Fama-French: Mkt + Size (SMB) + Value (HML)

1997
Carhart 4F
Momentum (WML) added; most robust cross-asset anomaly

2015
FF 5-Factor
Profitability (RMW) + Investment (CMA) factors added

2020s
AI + Alt Data
ML, NLP, non-linear models, unstructured alternative data

Today: Multi-factor models combine value, momentum, quality, size, and alternative signals using both linear and non-linear methods across asset classes.

Factor Categories

Value
估值

EP, BP,
Dividend Yield

Growth
成长

Revenue growth,
EPS growth

Momentum
动量

6M/12M return,
analyst revision

Quality
盈利质量

ROE, cash flow,
earnings stability

Low Vol
低波

Realized vol,
downside risk

Liquidity
流动性

Turnover,
bid-ask spread

$r_j = a_j + b_{j1}F_1 + b_{j2}F_2 + \ldots + b_{jn}F_n + \varepsilon_j$ Stock return explained by multiple factor exposures

Multi-Factor Models & Consistent Returns

Why Multi-Factor?

1. **Noise Reduction**
   Uncorrelated factors cancel each other's noise, stabilizing alpha

2. **Broader Premia Capture**
   Value, momentum, and quality premia are cyclical — blending is more consistent

3. **Higher IC Stability**
   Multi-factor IC is more stable over time than any single factor

4. **Better Risk-Adjusted Returns**
   Lower drawdowns and tracking error vs. single-factor strategies

The Fundamental Law of Active Management

$$\text{Alpha} = \text{IC} \times \sqrt{\text{Breadth}} \times \text{TC} \times \text{Risk}$$

IC

Information Coefficient

Forecast accuracy: how well your signal predicts returns

√B

Breadth

Number of independent bets: more stocks, more decisions

TC

Transfer Coefficient

Efficiency of translating signals into portfolio positions

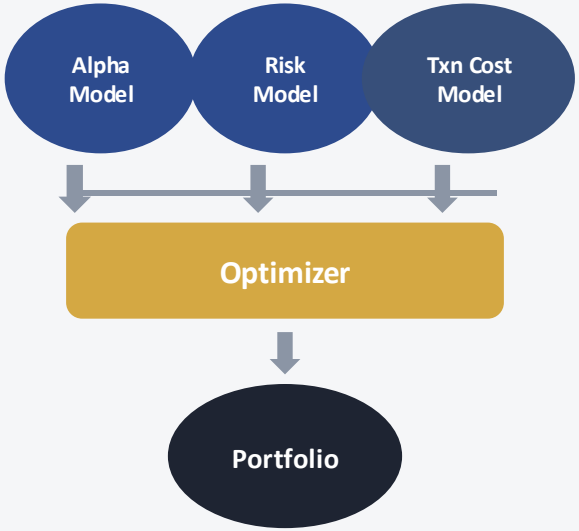
σ

Risk

Level of active risk budget you are willing to take

Multi-Factor Model Framework

Model Architecture



Research Process

① Factor Research

Data analysis · Factor development
Factor backtesting

② Strategy Construction

Objective setting · Factor combination
Strategy backtesting

③ Strategy Execution

Portfolio optimization · Trade execution
Data processing · Factor calculation

④ Performance Analysis

Return calculation · Attribution
Risk exposure · Model refinement

From Passive to Active: Product Forms

Passive Index	Smart Beta	Index Enhanced	Active Quant
Low cost, rules-based	← Less active	More active →	Alpha-seeking, flexible

Smart Beta

β+

systematic beta

improvement via factor tilts

Index Enhancement

β + α

benchmark-tracking

with alpha overlay

Active Quant

β + α++

unconstrained stock

selection for higher α

Quant Hedge

Pure α

market-neutral;

beta fully hedged

Machine Learning and AI in Factor Investing

DATA SOURCES

Financial Data

FEATURE ENG.

FACTORS

MODELING

OUTPUT

B

Project Brief & Technical Framework

Building the strategy step by step

“Barbell” Strategy

Single-style strategies face prolonged drawdowns. The barbell combines two extremes for more robust performance.

DEFENSIVE

Dividend Yield
Low Volatility
Quality / ROE
Earnings Stability

BARBELL

AGGRESSIVE

Small-Cap Tilt
Momentum
Earnings Growth
Analyst Revision

Research Process: Five Stages

1 Research

Literature survey; index methodology; market landscape

2 Data

Source, clean, validate; PIT alignment; EDA

3 Factor

Construct, standardize, neutralize; single-factor testing

4 Portfolio

Composite scoring; barbell allocation; weighting scheme

5 Backtest

Performance metrics; attribution; robustness; sensitivity

Step 1: Define the Stock Universe for Each Smart Beta Portfolio

CSI All-Share	5,000+ stocks
Liquidity Filter	~4,000 stocks
Exclude ST / *ST	~3,800 stocks
Exclude New IPOs	~3,600 stocks

Discussion: Why exclude ST stocks and recent IPOs? How does the universe choice affect factor exposure and strategy capacity?

Step 2: Factor Construction Pipeline

Raw Calc	Winsorize	Z-Score	Neutralize	Composite
Compute factor value for each stock (e.g., EP = E_TTM / MktCap)	Cap extremes at ±3σ to prevent outlier distortion	Standardize cross-sectionally: $Z = (F - \mu) / \sigma$	Regress out industry and size biases; use residuals	Combine multiple factors per barbell end

Neutralization Formula: $F_neutralized = F_raw - (\beta_1 \cdot Industry + \beta_2 \cdot \ln(MktCap) + \beta_3 \cdot Risk)$

Purpose: Remove sector and size biases to isolate the pure factor signal

Critical Concept: Point-in-Time Data

Look-Ahead Bias: Using future information that was not available at the time of the investment decision. This produces unrealistic backtest results.

Report Period End	Announce Date	Net Profit	Usable After
2024/12/31	2025/4/3	￥150M	Apr 3, 2025
2025/3/31	2025/4/27	￥155M (Q1)	Apr 27, 2025
2025/6/30	2025/8/8	￥310M (H1)	Aug 8, 2025
2025/9/30	2025/10/25	￥480M (9M)	Oct 25, 2025

A-Share Financial Reporting Calendar

Annual Report: by Apr 30 next year | **Q1:** by Apr 30 | **Interim:** by Aug 31 | **Q3:** by Oct 31

Calendar effect: Factor signals are strongest right after earnings seasons (April–May, August, October–November)

Step 3: Single-Factor Testing

Quantile-Sort Backtest

1. Rank all stocks by factor value at each rebalancing date

2. Sort into 5 (quintiles) or 10 (deciles) equal groups

3. Compute equal-weighted return for each group

4. Long–short spread = Top group − Bottom group

5. Plot cumulative returns; verify monotonic spread across groups

Key Evaluation Metrics

IC — Spearman rank correlation between factor and next-period return

ICIR — Mean(IC) / Std(IC); >0.5 meaningful, >1.0 strong signal

Turnover — % of portfolio changed at rebalance; drives transaction costs

Monotonicity — Top quantile should outperform bottom; spread should be consistent

Step 4: Smart Beta Portfolio Construction

Construction Steps

1. Normalize each factor (cross-sectional z-score)

2. Build composite score: weighted average of factor z-scores

3. Select top-N stocks by composite score

4. Weight by equal or score-proportional scheme

5. Rebalance monthly; track vs. CSI 300 benchmark

Key Design Choices

1. **Factor Weighting**
   Equal-weight vs. IC-weighted across selected factors

2. **Stock Count (N)**
   Top 50 = concentrated; Top 100 = diversified, lower TE

3. **Rebalance Freq.**
   Monthly recommended; quarterly reduces turnover cost

4. **Sector Constraint**
   Neutral limits concentration; unconstrained allows bigger bets

Step 5: Barbell Portfolio Construction & Evaluation

Portfolio Construction

- Select top-N stocks per composite score
- Allocation: 50/50 static vs. risk-parity
- Weighting: equal vs. market-cap
- Rebalance: monthly recommended

Performance Metrics

- Annualized return, volatility, Sharpe ratio
- Maximum drawdown, Calmar ratio
- Active return, tracking error, IR vs. bench
- Defensive vs. Aggressive attribution

Robustness & Sensitivity Checks

Sub-period analysis
(bull / bear / crisis)

Parameter sensitivity
(# stocks, frequency)

Sector-neutral vs.
unconstrained

Out-of-sample
holdout test

C

Next Steps

Schedule, deliverables, and what to do this week

Project Schedule & Milestones

Session	Date	Format	Key Focus
★ Intro (Today)	**Mar 15 (Sun)**	**Onsite**	**Quant overview; project brief;**
Meeting 1	Mar 22 (Sun)	Online	Choose 2 strategies; find benchmarks; read index docs; define factors & data; EDA
Meeting 2	Mar 29 (Sun)	Online	Single-factor backtests; allocation design
Meeting 3	Apr 12 (Sun)	Online	Full strategy review; robustness; presentation prep
★ Final	**Apr 19 (Sun)**	**Onsite**	**Team presentations (20 min + 10 min Q&A)**

10:00–11:00 for online meetings | All sessions on Sundays

Deliverables & Grading

Deliverables

- **Strategy Proposal**
  One-page write-up · due before Progress Meeting 1
- **Data Analysis & Factor Report**
  Factor analysis and data exploration · due before Progress Meeting 2
- **Backtest Notebook**
  Documented Python notebook · due before Progress Meeting 3
- **Final Report**
  10 to 15 pages · due on the final day
- **Presentation**
  15 to 30 slides · 20-minute talk with Q&A
- **Code Repository**
  Clean, documented, and reproducible code

Assessment Criteria

All of the following will be considered in grading.

Strategy Understanding
Clarity of rationale for chosen smart beta strategies

Factor Design
Quality and rigor of factor definitions and data choices

Analytical Depth
Thoroughness of data exploration and factor analysis

Backtest Quality
Soundness of methodology and interpretation of results

Presentation and Writing
Clarity, structure, and professional quality of all outputs

Code Quality
Readability, documentation, and reproducibility of code

This Week: Your First Assignment

Complete all 7 tasks before **Meeting 1 (Mar 22)** · Present your findings in the first progress meeting

#	TASK	
1	**Choose Two Smart Beta Strategies**	Decide which two factor strategies your team will build (e.g. Dividend + Microcap, Value + Momentum, Quality + Low-Vol). Document your rationale.
2	**Find Similar Real-World Indexes**	Identify at least one published index for each strategy (e.g. MSCI Value, CSI Dividend). Record full name, provider, and ticker.
3	**Read Index Documentation**	Download and read the methodology document for each index. Understand its universe, factors used, weighting scheme, and rebalance rules.
4	**Understand Index Construction**	Summarise how each index selects and weights stocks. Note any screens (liquidity, sector caps) and how often it rebalances.
5	**Define the Factors You Will Use**	Write out each factor formula (e.g. P/B ratio, 12-1 month return). State the expected direction and economic intuition.
6	**Identify Data Needed**	List every data field required to compute your factors (price, financial statement items, dates). Note frequency and source.
7	**Get Familiar with the Data**	Load the provided dataset. Check coverage, missing values, and data types. Run basic descriptive stats on key fields.

Questions & Discussion

Data package will be distributed after this session

Strategy Proposal due before Meeting 1 (Mar. 22)

AI tools are encouraged — but you must understand and defend your work